Digital innovation diffusion in the manufacturer–distributor relationship

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Abstract

Purpose – In the era of digital transformation, digital innovation has emerged as a crucial driver of competitiveness and growth. However, limited knowledge exists on how digital innovation diffuses across supply chains, particularly in the manufacturer–distributor relationship. Drawing upon diffusion of innovation theory and social network theory, this study aims to examine how manufacturers' digital innovation impacts distributors' digital innovation, focusing on the moderating roles of cooperation length and network status of manufacturers and distributors.

Design/methodology/approach – The study uses a sample of 233 manufacturer–distributor pairs from the Chinese paper-making industry. Geographically, the survey comprises sample firms nationwide, covering most regions and provinces in China.

Findings – The authors find that manufacturers' digital innovation positively affects distributors' digital innovation. This relationship is strengthened by cooperation length and manufacturers' network status but dampened by distributors' network status. The findings provide insights into how digital innovation diffuses across supply chain partners and highlight the importance of social relationships and network positions in this digital innovation diffusion process.

Originality/value – This study makes a significant contribution to the field of supply chain management by offering deep insights into the diffusion of digital innovation across supply chain partners. It highlights the significance of social relationships and network positions in the process of digital innovation diffusion, offering a novel perspective on the interplay between manufacturers and distributors in the context of digital transformation.

Keywords Digital innovation, Innovation diffusion, Manufacturer–distributor relationship, Cooperation length, Network status

Paper type Research paper

Introduction

In the era of digital transformation, digital innovation has emerged as a pivotal driver of competitiveness and growth for firms across industries (Greenan and Napolitano, 2024; Kumar *et al.*, 2018; Lerman *et al.*, 2022; Nambisan *et al.*, 2017; Spieth *et al.*, 2021). Defined as the utilization of digital technology to develop new business models, products or services, with the aim of disrupting existing markets or pioneering new ones, digital innovation is critical in enabling firms to respond to market changes, enhance customer experiences and streamline operations (Di Vaio *et al.*, 2021; Fichman *et al.*, 2014). The importance of digital innovation is underscored by its potential to create new business opportunities, usher in groundbreaking advancements in multiple industries and contribute to overall economic development (Nambisan *et al.*, 2017; Vial, 2019). Research on digital innovation focuses on its antecedents, including market level (Nambisan *et al.*, 2017; Svahn *et al.*, 2017), firm level (Matt *et al.*, 2015; Van Looy, 2021) and managerial level (Ferreira *et al.*, 2019; Firk *et al.*, 2022) that enable successful implementation, as well as its impact on competitive advantage, operational efficiency and market positioning (Fichman *et al.*, 2014; Porter and Heppelmann, 2014), highlighting its role as a catalyst for business transformation and market adaptability (Belhadi *et al.*, 2022; Liu *et al.*, 2023).

Despite the vast research on digital innovation, a significant limitation remains in our comprehension of how these advancements diffuse across organizations. Innovation diffusion pertains to the process through which new ideas, technologies or practices spread among individuals, organizations or markets (Ho, 2022; Moraes *et al.*, 2025). Diffusion of innovation theory, a robust framework for understanding the spread of new ideas and technologies, provides valuable insights but faces unique challenges when applied to digital innovation (Zanello *et al.*, 2016; Zhang and Vorobeychik, 2019). In the context of supply chains, understanding digital innovation diffusion, referring specifically to the process through which digital innovations are transmitted from manufacturers to distributors, becomes even more pivotal. According to the KPMG Global Tech Report 2023 (KPMG, 2023), digital transformation and digital innovation in the manufacturing industries are receiving increasing attention, with a particular focus on whether its sales entities can leverage digital innovation technologies for enhancement. In particular, exploring the diffusion of digital innovation from manufacturers to

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distributors promises to shed light on how firms can collaborate and learn from each other to drive innovation and growth within the supply chain (Alinaghian *et al.*, 2021; Xu *et al.*, 2023). By leveraging diffusion of innovation theory and accounting for the unique dynamics and complexities of digital advancements, we can gain a more comprehensive understanding of how digital innovation spread.

In the process of digital innovation diffusion within the manufacturer–distributor relationship, managing supply chain risks emerges as a critical challenge. Supply chain risks, which include supply disruptions, demand fluctuations, logistical issues, financial instability and regulatory shifts (Um and Han, 2021; Wang *et al.*, 2024), pose significant threats to the seamless flow of digital innovations. When faced with these risks, the importance of supply chain relationships cannot be overstated and the role of social influence is equally significant (Ho *et al.*, 2015; Lu *et al.*, 2018). Strong, trust-based relationships facilitate not only better communication and collaboration but also harness the power of social influence to mitigate risks. When manufacturers and distributors share mutual understanding and trust, they are more likely to cooperate effectively, share crucial information promptly and adapt flexibly to unforeseen challenges. This social dimension of supply chain relationships, rooted in mutual respect and shared goals, becomes a vital asset in overcoming the complex risks associated with digital innovation diffusion. By leveraging these relationships and the inherent social influence they carry, firms can enhance their resilience and ensure the successful adoption and spread of digital innovations throughout the supply chain.

In particular, cooperation length, referring to the duration of the business relationship between a manufacturer and its distributor, serves as a proxy for the depth and stability of social relationships between manufacturers and distributors (Han *et al.*, 2022; Wang *et al.*, 2015). A longer cooperation period fosters deeper trust, enabling the seamless exchange of information, knowledge and resources (Gao *et al.*, 2015; Wang *et al.*, 2015), which, in turn, accelerates the diffusion of digital innovation. On the other hand, network status reflects the social influence of a firm within its network of relationships, which determines its access to information, resources and influence (Kim and Zhu, 2018; Wang *et al.*, 2019). Status power underscores the fact that the focal firm possesses a closer proximity to all other network members, thereby exerting significant influence over them within the networks (Lan *et al.*, 2020; Wang *et al.*, 2019). By considering the moderating effects of cooperation length and network status of manufacturers and distributors, we can gain a more comprehensive understanding of how digital innovation diffuses in the manufacturer–distributor relationship. This understanding provides valuable insights for firms seeking to leverage digital technologies to enhance their supply chain efficiency and overall competitiveness.

To surmount these limitations, our research aims to investigate the following specific research questions:

RQ1. How does manufacturers' digital innovation impact distributors' digital innovation, and how are these effects moderated by cooperation length, manufacturers' network status and distributors' network status?

By answering these questions, we seek to provide insights into the mechanisms through which digital innovations diffuse within the supply chains and how firms can leverage these mechanisms to enhance their digital transformation and competitiveness. To achieve this purpose, we use a sample of 233 pairs of manufacturers and distributors in the Chinese paper-making industry. As a traditional manufacturing sector facing an urgent need for digital transformation to enhance efficiency, this industry presents a unique opportunity to observe the spread of technological innovations within complex supply chains. In particular, it enables us to closely examine the diffusion of digital innovation among supply chain partners, providing crucial insights into the ongoing digitalization process in traditional manufacturing.

This study offers several significant contributions. First, it advances our understanding of how digital innovation diffuses in the manufacturer–distributor relationship. Drawing on the diffusion of innovation theory, this study examines the mechanisms through which manufacturers' digital innovation impacts distributors' digital innovation. This theoretical framework provides insights into the complex interplay between upstream and downstream firms in the supply chain, thereby enriching and expanding existing knowledge on digital innovation and its application in supply chain management. Second, this study extends the application of social network theory to the realm of digital innovation diffusion in supply chains. By incorporating cooperation length as a moderator, the research highlights the importance of relationship stability and trust in facilitating the exchange of knowledge and resources in the supply chain. This theoretical extension offers new insights into how cooperation length shapes the diffusion process of digital advancements across supply chain partners. Third, this study explores the moderating roles of manufacturers' and distributors' network status, significantly enhancing our understanding of how firms' strategic positions within their networks influence digital innovation diffusion in the supply chains. Drawing upon social network theory, this study examines how manufacturers' and distributors' network status differently moderate their capacity to adopt and disseminate digital advancements. This theoretical integration offers a comprehensive view of how network dynamics moderate the diffusion of digital innovation across supply chain partners, thereby extending and enriching existing theories on innovation diffusion, social networks and their intersection in supply chain management.

Theoretical framework and hypotheses development

Digital innovation

In today's rapidly evolving business landscape, digital innovation has emerged as a crucial driver for corporate survival and growth (Kohli and Melville, 2019; Mithas *et al.*, 2022). It serves not only as a source of new value propositions and revenue streams but also as a pivotal tool for businesses to swiftly adapt to changing market dynamics and consumer preferences (Liu *et al.*, 2021; Spieth *et al.*, 2021). By harnessing advanced technologies such as artificial intelligence, big data analytics and the Internet of Things, digital innovation propels firms toward a more intelligent, interconnected and efficient future (Di Vaio *et al.*, 2021; Fichman *et al.*, 2014). Digital innovation not only helps firms differentiate themselves from competitors but also streamlines operations and enhances customer experiences, thereby strengthening their core competitiveness (Nambisan *et al.*, 2017; Vial, 2019). In the digital economy, digital innovation has transcended mere technological trends and become a strategic imperative for businesses seeking to remain competitive in the new century. Through digital innovation, firms can break free from traditional industry constraints, explore new market potentials and achieve leapfrog development (Fichman *et al.*, 2014; Hanelt *et al.*, 2021).

The exploration of digital innovation in academic research has revealed two profound trends. The first trend delves into identifying the antecedents that enable the successful implementation of digital innovation, encompassing a wide range of factors (Ferreira *et al.*, 2019; Van Looy, 2021). At the market level, technological advancements, market dynamics and competitive pressures are integral components. Technological advancements, such as the availability of cutting-edge technologies, provide the foundation for digital innovation (Nambisan *et al.*, 2017). Market dynamics, including product diversity on multilateral platforms (Joglekar *et al.*, 2022), shape the opportunities and challenges for digital initiatives. Competitive pressures force firms to continuously innovate to stay ahead of rivals (Xie and Wu, 2024). At the firm level, strategic orientations (Matt *et al.*, 2015) and organizational culture (Isensee *et al.*, 2020) play a crucial role in determining the success of digital innovation. Top management support and a culture that encourages experimentation and risk-taking are essential for fostering a digital mindset within the organization. From a managerial perspective, personal characteristics of managers, such as gender, age and education (Ferreira *et al.*, 2019) influence their approach toward digital innovation, which ultimately impacts its implementation. Managers with a forward-thinking mindset and a strong technical background are better positioned to lead digital innovation efforts within their organizations (Ferreira *et al.*, 2019).

The second trend focuses on the impacts of digital innovation (Kohli and Melville, 2019; Svahn *et al.*, 2017). This trend delves into how digital innovation empowers companies to maintain sustainable competitiveness, optimizing their operations and market positioning (Liu *et al.*, 2023; Vial, 2019). Digital innovation revolutionizes innovation processes by introducing novel methodologies and tools that expedite product development cycles and elevate product quality standards. By dismantling traditional silos and fostering cross-functional collaboration, it reshapes organizational forms, fostering more informed decision-making and elevating operational effectiveness (Fichman *et al.*, 2014; Kohli and Melville, 2019). These operational gains manifest in improved productivity and streamlined workflows, resulting in cost reductions and optimized resource utilization (Kalesh *et al.*, 2024; Porter and Heppelmann, 2014).

Digital innovation diffusion

However, the existing research on digital innovation has notable gaps in exploring its diffusion, particularly within the supply chain. The expansion of digital innovation across the various stages of the supply chain is paramount as it determines the extent to which the benefits of these innovations can be realized (Palm, 2022; Yin *et al.*, 2022). The lack of sufficient inquiry into digital innovation diffusion limits our understanding of how these innovations propagate through supply chains, affecting not only the initial innovators but also the entire ecosystem of stakeholders. Understanding the mechanisms and dynamics of digital innovation diffusion is crucial for firms seeking to maximize the impact of their investments in digital technologies.

The diffusion of innovation theory provides a framework for understanding how new ideas, technologies or practices spread through social systems (Shao *et al.*, 2025; Vargo *et al.*, 2020). This theory explores how information is transmitted to individuals or organizations over time and how this process facilitates the application of innovation (Alinaghian *et al.*, 2021; Bankel and Govik, 2024). The theory also identifies six key characteristics of innovations that influence their diffusion: relative advantage, image, compatibility, complexity, trialability and observability (Yuen *et al.*, 2021; Zanello *et al.*, 2016). By applying this theory to digital innovation diffusion, we can explore the mechanisms through which digital technologies and solutions propagate across industries, pinpoint factors that accelerate or impede their broad adoption and gain valuable insights for predicting the success of digital innovations and devising effective implementation strategies (Ho, 2022; Wang *et al.*, 2024).

Specifically, examining the diffusion of digital innovation in the manufacturer–distributor relationship is paramount. According to the diffusion of innovation theory, the adoption and dissemination of innovations adhere to distinct patterns within social systems (Alinaghian *et al.*, 2021; Vargo *et al.*, 2020). In the supply chain context, manufacturers, as upstream entities, not only enhance their own operational efficiency and market competitiveness through digital innovation but also significantly influence downstream distributors. This relationship offers a distinctive perspective on how digital technologies transform operational dynamics and strategic alignment between upstream manufacturers and downstream distributors. Furthermore, analyzing this diffusion process sheds light on how distributors use digital innovation to strengthen their market positioning, operational efficiency and overall business performance.

In exploring the diffusion of digital innovation within the manufacturer–distributor relationship, a crucial aspect that cannot be overlooked is supply chain risks. Supply chain risks refer to the various uncertainties and potential losses that may arise during the operation of the supply chain, significantly impacting its efficiency and effectiveness, which can often be classified into internal risks (e.g. process and control risks), network-related risks and external risks (e.g. COVID-19 and political instability) (Ho *et al.*, 2015; Um and Han, 2021).

In the context of digital innovation diffusion, these risks present new characteristics and challenges. Such risks stem primarily from the instability and uncertainty in the cooperative relationships between manufacturers and distributors. Due to changes in market conditions, adjustments in business strategies or information asymmetry, the cooperative relationships between manufacturers and distributors may be disrupted (Colicchia and Strozzi, 2012; Um and Han, 2021), subsequently affecting the smooth diffusion of digital innovation. To mitigate these risks and ensure that digital innovation can effectively propagate and generate value within the supply chain, social relationships and social influence become particularly important. Such an analysis provides a more comprehensive perspective for understanding digital innovation diffusion within the supply chain and offers valuable guidance for businesses on how to effectively address supply chain risks and promote digital innovation in practice.

To address the existing gaps in the understanding of digital innovation diffusion within the supply chains (Alinaghian *et al.*, 2021; Wang *et al.*, 2024), our research delves deeper into the intricate dynamics from manufacturers to distributors. Guided by social network theory, we aim to analyze how manufacturers' digital innovation influences distributors' digital innovation while uncovering the nuanced roles of social relationships (measured by cooperation length) and social influence (measured by manufacturers' and distributors' network status) as moderating factors. This exploration not only fills a crucial void in the literature but also equips businesses with actionable insights to navigate the complexities of digital innovation diffusion (van Hoek, 2024; Vargo *et al.*, 2020). Through this comprehensive lens, we strive to enable firms to harness the full potential of digital technologies, foster resilient supply chain relationships and mitigate associated risks, ultimately driving sustainable competitiveness and growth in the digital era.

Digital innovation diffusion in the manufacturer–distributor relationship

In investigating digital innovation diffusion in the manufacturer–distributor relationship, three primary mechanisms emerge as pivotal. First, as the upstream segment of the supply chain, manufacturers' innovation practices and technology adoption often set the technological standards and operational paradigms for the entire supply chain (Dong *et al.*, 2020; Kohli and Melville, 2019). When manufacturers adopt and implement digital innovation, these new technologies, methodologies or standards are likely to become the benchmarks that subsequent supply chain segments, including distributors, need to follow or adapt to.

Second, manufacturers' digital innovation activities are often accompanied by changes in market demand and shifts in consumer preferences (Di Vaio *et al.*, 2021; Nambisan *et al.*, 2017). These changes not only affect manufacturers' products and services but also influence distributors' market strategies and operational decisions. To maintain alignment with manufacturers and satisfy market demand, distributors may need to keep pace with manufacturers' digital innovation efforts.

Third, digital innovation often requires alignment and integration across supply chain partners (Fichman *et al.*, 2014; Vargo *et al.*, 2020). Manufacturers' digital innovation often involves new systems, processes and standards that distributors need to adapt to maintain efficient operations. This alignment pressure pushes distributors to innovate digitally to ensure compatibility and interoperability with manufacturers. As a result, manufacturers' digital innovation often drives distributors' digital innovation.

Therefore, we suggest:

H1. Manufacturers' digital innovation is positively associated with distributors' digital innovation.

The moderating role of cooperation length and network status

In the context of digital innovation diffusion between manufacturers and distributors, the moderating role of social relationships and social influence emerges as the crucial consideration. Specifically, cooperation length, reflecting social relationships, captures the duration and depth of the relationship between manufacturers and distributors (Han *et al.*, 2022; Lu *et al.*, 2018), while network status, reflecting social influence, indicates their positions and influence within the network (Kim and Zhu, 2018; Wang *et al.*, 2019). These factors moderate the diffusion process by shaping the channels of communication, levels of trust and access to resources that facilitate or hinder the spread of digital innovation.

Cooperation length refers to the duration of the partnership between manufacturers and distributors (Gao *et al.*, 2015; Han *et al.*, 2022). This variable is significant as a moderator in digital innovation diffusion because it captures the extent to which the parties have built trust, familiarity and shared understanding through their collaborative efforts. Longer cooperation lengths indicate a more established relationship, which may enhance the willingness and ability of distributors to adopt digital innovation (Han *et al.*, 2022; Zhao *et al.*, 2008). Conversely, shorter cooperation lengths may hinder diffusion due to the lack of trust and familiarity, requiring manufacturers to invest more in persuasion to overcome barriers to adoption (Lu *et al.*, 2018; Wang *et al.*, 2015).

Network status reflects the position and hierarchical advantage of the network (Lan *et al.*, 2020; Wang *et al.*, 2019). Both parties' network status plays a moderating role in digital innovation diffusion. For manufacturers, high network status indicates the central position within the supply chain, providing greater access to resources, information and potential partners. This status enhances their ability to disseminate digital innovation effectively and influence distributors' adoption decisions. Conversely, low network status may limit manufacturers' influence and require them to rely more on other mechanisms, such as incentives or coercion, to promote diffusion. Similarly, distributors' network status affects their access to innovative information, their willingness to experiment and their ability to influence other members of the supply chain. High-status distributors have stronger bargaining power, while low-status distributors are more likely to comply.

Cooperation length plays a crucial role in enhancing the impact of manufacturers' digital innovation on distributors' digital innovation in several ways. First, as the cooperation length increases, the parties develop deeper trust and familiarity with each other (Han *et al.*, 2022; Zhao *et al.*, 2008). This enhanced trust and familiarity facilitate the smooth transmission of information, including digital innovation practices and technologies. Distributors are more likely to adopt digital innovation introduced by manufacturers with whom they have a long-standing and trusted relationship. The trust built over time reduces distributors' perceived risks associated with adopting new technologies, making them more receptive to manufacturers' digital innovation initiatives.

Second, longer cooperation lengths allow manufacturers and distributors to develop a shared understanding of business operations, market dynamics and customer needs (Han *et al.*, 2022; Wang *et al.*, 2015). This shared understanding enables distributors to better comprehend the rationale behind manufacturers' digital innovation decisions and the potential benefits for their own businesses. Furthermore, distributors can provide valuable feedback to manufacturers based on their market insights, which can further refine and tailor digital innovation to meet specific market demands. The alignment of expectations and goals fosters mutually beneficial partnerships, encouraging distributors to adopt and adapt digital innovation.

Finally, manufacturers with longer-term cooperative relationships with distributors are more likely to invest in relationship maintenance and enhancement (Gao *et al.*, 2015; Lu *et al.*, 2018). This investment can take the form of shared resources, training programs and joint innovation initiatives. Such investments not only strengthen the partnership but also provide distributors with the necessary capabilities and resources to adopt and implement manufacturers' digital innovations. Distributors are more inclined to reciprocate manufacturers' efforts by adopting new digital technologies and practices that enhance their own operational efficiency and market competitiveness. Thus, cooperation length enhances the positive relationship between manufacturers' digital innovation and distributors' digital innovation by facilitating mutual investment and support in relationship maintenance and enhancement.

Therefore, we suggest:

H2. Cooperation length positively moderates the positive relationship between manufacturers' digital innovation and distributors' digital innovation.

Manufacturers' network status also plays a pivotal role in boosting the effectiveness of manufacturers' digital innovation in driving distributors' digital innovation in numerous ways. First, manufacturers with high network status enjoy greater visibility and credibility (Kim and Zhu, 2018; Lan *et al.*, 2020). Their digital innovation and practices are more likely to attract attention and influence other supply chain members, including distributors. When manufacturers with high network status introduce digital innovation, distributors are more likely to perceive these innovations as valuable and trustworthy. The manufacturers' reputation and standing in the network facilitate the diffusion of digital innovation to distributors, who are motivated to adopt them due to the perceived credibility of the sources.

Second, high network status often translates into access to a wider range of resources and information (Bellamy *et al.*, 2014; Wang *et al.*, 2019). Manufacturers with a strong network position are more likely to have access to cutting-edge technologies, market insights and best practices. This access enables them to develop and implement innovative digital solutions that are ahead of the curve. When distributors see manufacturers with high network status leading the way in digital innovation, they are more inclined to follow suit and adopt these innovative practices. The manufacturers' ability to share resources and knowledge with distributors further enhances the diffusion of digital innovation.

Third, manufacturers with high network status exert significant influence on other network members (Lan *et al.*, 2020; Wang *et al.*, 2019). Their opinions, decisions and actions are often followed and emulated. When manufacturers with high network status embrace digital innovation, their leadership and influence motivate distributors to do the same. Distributors see the manufacturers as role models and leaders in digital transformation, encouraging them to align their own digital innovation efforts with those of the manufacturers. This alignment not only strengthens the supply chain partnership but also positions both parties for success in the digital economy.

Therefore, we suggest:

H3a. Manufacturers' network status positively moderates the positive relationship between manufacturers' digital innovation and distributors' digital innovation.

Distributors' network status plays a pivotal role in dampening the positive relationship between manufacturers' digital innovation and distributors' digital innovation in several crucial ways. First, distributors with high network status may be more dependent on their existing business models, relationships and market positions (Lan *et al.*, 2020; Wang *et al.*, 2015). This dependency can lead to resistance to change, including the adoption of new digital innovation introduced by manufacturers. Distributors with high network status may perceive themselves as leaders in their own right and may be reluctant to follow manufacturers' lead in digital innovation. They may fear that adopting new technologies and practices could disrupt their existing business operations and relationships.

Second, distributors with high network status tend to have more autonomy in making business decisions (Wang *et al.*, 2015; Wang *et al.*, 2019). They may have established their own successful business models and strategies and may be less inclined to adopt manufacturers' digital innovation unless they see clear and compelling benefits for their own businesses. The autonomy of high-status distributors allows them to evaluate manufacturers' digital innovation independently and may result in more cautious or selective approaches to adoption.

Third, distributors who hold prominent network status may see manufacturers' digital innovation as a possible threat to their current market influence and sphere of operation (Porter and Heppelmann, 2014; Wang *et al.*, 2019). They could be concerned that embracing these innovations might result in altered market dynamics or affect their existing market position. Importantly, while distributors and manufacturers operate in complementary roles within the supply chain rather than as direct competitors, this territorial and competitive mindset could still hinder the seamless integration of manufacturers' digital advancements into distributors' operations. This occurs as distributors may be cautious about adopting innovations that might shift the balance of their established market presence.

Therefore, we suggest:

H3b. Distributors' network status negatively moderates the positive relationship between manufacturers' digital innovation and distributors' digital innovation.

Method

Sample and data collection

To test our hypotheses, we commenced data collection in 2020, focusing on the Chinese paper-making industry. The Chinese paper-making industry boasts a long history and solid industrial foundation, but with the tightening of environmental policies and the intensification of market competition, the industry is facing pressure to transform and upgrade. Digital innovation has emerged as a crucial path for the industry to enhance competitiveness, optimize production processes and reduce costs. Therefore, studying the diffusion of digital innovation, especially from manufacturers to distributors, in the Chinese paper-making industry is significant for understanding the digital transformation of traditional manufacturing industries.

Since our data collection focused on the dyadic relationship between manufacturers and distributors, we designed two paired questionnaires: one for manufacturers and another for distributors. We based the design of these questionnaires on related literature (Li *et al.*, 2011; Osmonbekov *et al.*, 2009). Given the scarcity of detailed information on dyadic cooperation between manufacturers and distributors in secondary sources, a direct survey approach became crucial to our research.

In designing the study, we carefully followed precedents set by prior research to operationalize our variables (Li *et al.*, 2011; Oehme and Bort, 2015). We conducted rigorous pretests to enhance the validity of our measures. Initially, we interviewed executives from two paper-making manufacturers and five distributors, gaining valuable insights into their perspectives on digital innovation and cooperation activities. These interviews proved indispensable in refining our survey questions, ensuring they captured the nuances of our research objectives. Subsequently, we conducted a pilot study involving another three paper-making manufacturers and ten distributors, interviewing executives from each of these firms. The feedback obtained from this pilot study was instrumental in further refining and finalizing our questionnaires, enhancing their reliability and relevance for our target firms. This rigorous methodology laid the foundation for our study. All manufacturers and distributors that participated in the interviews and the pilot survey were excluded from the formal survey.

Data collection was carried out following a two-phase procedure. First, we contacted an important industry alliance consisting of 25 important Chinese paper-making manufacturers, which collectively have over 1,200 distributors. To ensure a comprehensive, representative and diverse sample, the alliance randomly selected 50% of the distributors, providing us with a list of 600. In 2020, we conducted a targeted mail survey, aiming to gather specific and nuanced data from senior executives of these distributors. We chose these executives as they possess in-depth knowledge of their organizations' overall operations, including digital innovation and cooperation activities with other organizations. If a distributor has multiple manufacturers, it is necessary for the distributor to complete several questionnaires, each tailored to capture the specifics of the relationship with a distinct manufacturer. These manufacturers can be among the original 25 or any other manufacturers the distributor collaborates with. After the initial mail survey, we implemented a series of reminders and follow-up phone calls to encourage participation and authenticate the responses received. With the invaluable support of the manufacturers, we were able to achieve a noteworthy 44.2% response rate, resulting in 265 completed surveys. This response rate compares favorably with those reported in prior studies (e.g. Valentine and Fleischman, 2004; Zahra, 2020). However, due to the presence of missing data on certain variables, 246 questionnaires were complete and acceptable for analysis. Second, we mailed questionnaires to the corresponding manufacturers. With the same follow-up and reminder procedures, we received 233 responses, all of which were complete and satisfactory, resulting in a response rate of 94.7%. Thus, we obtained a final sample of 233 pairs of manufacturers and distributors. Geographically, our survey comprises sample firms nationwide, covering most regions and provinces in China.

Measures

Independent variable and dependent variable

Manufacturers' digital innovation and distributors' digital innovation. We measured digital innovation by using patents that intensely leverage digital technologies (Liu *et al.*, 2021; Nambisan *et al.*, 2017). Patents serve as a widely accepted metric for innovation among scholars, primarily due to the thorough and standardized patent application examination process. This ensures a systematic representation of a firm's innovative endeavors (Hanelt *et al.*, 2021; Zahra and Nielsen, 2002). We sourced comprehensive data on various patents from the SIPO database. Two groups of researchers coded independently whether each patent significantly incorporated digital technologies. This evaluation involved scrutinizing the patent descriptions for the inclusion of digital elements. If a patent exhibited the use of such digital technologies, it was counted as one digital innovation for that particular year. Subsequently, we tallied the number of patents in each year to determine digital innovation. To ensure the accuracy of our coding methodology, we compared the findings from both groups of researchers and resolved any discrepancies through discussion until a consensus was reached.

We used distributors' digital patent data from 2019 as a proxy to measure distributors' digital innovation. Similarly, to account for the time lag, we used manufacturers' digital patent data from 2018 as a proxy to measure manufacturers' digital innovation.

Moderating variables

Cooperation length. We used the duration (years) of the relationship between the matched pair of manufacturers and distributors to measure cooperation length.

Manufacturers' network status and distributors' network status. We adopted degree centrality to measure network status (Oehme and Bort, 2015; Wang *et al.*, 2019).

During the survey, we used the name generator method to collect data on social networks (Batjargal *et al.*, 2013; Burt, 1992). The name generator method offers a thorough assessment of network structural properties, reducing the potential for social desirability bias when compared to methods like the position generator. Taking into account the time lag, senior executives were requested to carefully compile a list up to 15 organizations, with whom their firms had significant collaborations (technical, business, etc.) over the past three years (2016–2018). Senior executives were also asked to answer one question:

Q1. Do these two organizations interact with each other?

By applying this approach across all sampled firms, we successfully mapped out the network relationships between the primary actor and its alternates, as well as the relationships among other alternates. Thus, we successfully obtained all firms' ego networks.

To ensure the validity of network relationships, we conducted additional verification checks. In addition to the subjective network data compiled by senior executives, we conducted data-gathering cooperative activities on Baidu (the largest and most authoritative search engine in China, similar to Google) (Bennett *et al.*, 2015; Kaul and Wu, 2016). To further enhance data accuracy, we cross-referenced the information obtained from Baidu with data from company websites and mainstream news sources (Li *et al.*, 2008). We thoroughly scoured these sources for details on contracts, transactions, strategic alliances, joint venture investments and other forms of cooperative activities. It is crucial to note that all data gathered from these additional sources covered the same time period (2016–2018) as our previous subjective network data, ensuring consistency and comparability. Finally, we verified that all objective network relationships are consistent.

Based on the ego networks of manufacturers and distributors, we obtained manufacturers' network status and distributors' network status.

Control variables. Following prior research, our analysis incorporated multiple control variables that theoretically and practically influence our model (Cinelli *et al.*, 2022). First, we controlled for firm age measured by the number of years since a firm's establishment because age. Moreover, we also controlled for firm size, measured by the logarithm of the number of employees. Both firm age and firm size may influence a firm's ability to leverage digital innovation (Liu *et al.*, 2023). Thus, we controlled for manufacturers' firm age, manufacturers' firm size, distributors' firm age and distributors' firm size.

Modeling

We test our hypotheses using a count data model because our dependent variable, distributors' digital innovation, is a count measure. Given that the dependent variable exhibits overdispersion, which is not well-suited for a Poisson model, we opted for a negative binomial model as our primary analytical approach.

Analysis and results

Table 1 provides the means, standard deviations and correlation coefficients for all variables. Table 1 also indicates that there are no significantly high correlations among the studied variables.

Regression results

We report the results of our negative binomial regression analysis in Table 2. Model 1 presents the base model on distributors' digital innovation. Model 2 shows the main effect of manufacturers' digital innovation on distributors' digital innovation. Models 3, 4 and 5 incorporate cooperation length, manufacturers' network status and distributors' network status, respectively, along with their corresponding interaction terms. Model 6 is the full model.

In Model 2, manufacturers' digital innovation is positively associated with distributors' digital innovation ($b = 0.353$, $p = 0.001$), which supports *H1*. A one-standard-deviation increase in manufacturers' digital innovation (i.e. 9.15) corresponds to a 42% increase of distributors' digital innovation. To put this 42% increase in perspective, suppose a distributor's current digital innovation is 3.95 digital patents – the mean of the dependent variable – and manufacturers' digital innovation is 9.15 (or 1 SD) higher than that of the current situation. All else being equal, we would then expect the distributor's digital innovation to increase by 1.66 digital patents, resulting in 5.61 digital patents.

As evident from Model 3, the interaction between manufacturers' digital innovation and cooperation length exhibits a statistically significant positive impact on distributors' digital innovation ($b = 0.433$, $p = 0.006$), providing support for *H2*. Model 4 reveals a statistically significant positive effect of the interaction between manufacturers' digital innovation and manufacturers' network status on distributors' digital innovation ($b = 0.326$, $p = 0.016$), which provides support for *H3a*. Model 5 indicates that the interaction of manufacturers' digital innovation and distributors' network status is statistically negative ($b = -0.239$, $p = 0.019$), supporting *H3b*. The results of the full model in Model 6 are consistent with the previously noted findings.

Robustness checks

To ensure the robustness of our findings, we conducted additional tests. First, we used an alternative measure for assessing the independent variable – manufacturers' digital innovation. Originally, we used 2018 data as our metric for manufacturers' digital innovation. Instead, we adopted an alternative approach, replacing this single-year metric with the average data from the preceding three years (2016–2018) to assess manufacturers' digital innovation. Models 7 and 8 in Table 3 present the results. Since the sample and the dependent variable remain unchanged, Model 1 in Table 2 is the base model. Model 7 shows the main

Table 1 Descriptive statistics and correlations ($n = 233$)

Variables	Mean	SD	1	2	3	4	5	6	7	8
1. Distributors' digital innovation	3.95	4.81								
2. Manufacturers' digital innovation	6.21	9.15	0.237							
3. Cooperation length	8.56	4.59	−0.169	−0.236						
4. Manufacturers' network status	10.33	3.53	−0.144	−0.203	0.088					
5. Distributors' network status	10.30	3.62	−0.041	0.009	0.067	−0.120				
6. Manufacturers' firm age	22.60	9.74	−0.019	−0.011	0.253	−0.179	0.006			
7. Manufacturers' firm size	6.75	0.90	0.077	−0.224	0.137	0.078	0.048	−0.227		
8. Distributors' firm age	17.04	7.62	−0.252	−0.014	0.439	0.079	0.085	0.045	0.055	
9. Distributors' firm size	3.54	0.87	0.211	−0.070	−0.074	0.030	−0.056	−0.031	0.008	−0.200

Source(s): Authors' own work

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Table 2 Test of hypotheses

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Manufacturers' firm age	0.000 (0.988)	0.006 (0.549)	0.010 (0.331)	0.005 (0.629)	0.007 (0.509)	0.011 (0.261)
Manufacturers' firm size	0.114 (0.308)	0.227* (0.045)	0.219† (0.052)	0.208† (0.065)	0.243* (0.029)	0.221* (0.046)
Distributors' firm age	−0.038** (0.004)	−0.039** (0.003)	−0.030* (0.042)	−0.039** (0.003)	−0.041** (0.002)	−0.033* (0.022)
Distributors' firm size	0.250* (0.036)	0.266* (0.023)	0.272* (0.018)	0.251* (0.030)	0.297* (0.011)	0.293** (0.009)
Manufacturers' digital innovation		0.353** (0.001)	0.486*** (0.000)	0.345** (0.001)	0.352** (0.001)	0.497*** (0.000)
Cooperation length			−0.047 (0.743)			−0.038 (0.783)
Manufacturers' digital innovation * cooperation length			0.433** (0.006)			0.501** (0.001)
Manufacturers' network status				−0.179† (0.076)		−0.168† (0.085)
Manufacturers' digital innovation * manufacturers' network status				0.326* (0.016)		0.382** (0.004)
Distributors' network status					0.092 (0.339)	0.111 (0.235)
Manufacturers' digital innovation * distributors' network status					−0.239* (0.019)	−0.271** (0.005)
Constant	0.291 (0.768)	−0.714 (0.475)	−0.864 (0.392)	−0.470 (0.641)	−0.930 (0.347)	−0.898 (0.372)
N	233	233	233	233	233	233
Log-likelihood	−551.037	−545.168	−540.226	−541.230	−542.088	−530.858

Note(s): p-values are in parentheses. †$p < 0.10$; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$
Source(s): Authors' own work

effect of manufacturers' digital innovation on distributors' digital innovation. Model 8 is the full model, which incorporates cooperation length, manufacturers' network status, distributors' network status and the interaction terms. Specifically, in Model 7, manufacturers' digital innovation is positively associated with distributors' digital innovation (b = 0.366, p = 0.001). Model 8 reveals a statistically significant positive effect of the interaction between manufacturers' digital innovation and cooperation length on distributors' digital innovation (b = 0.483 p = 0.001), the statistically significant positive effect of the interaction between manufacturers' digital innovation and manufacturers' network status on distributors' digital innovation (b = 0.390, p = 0.002) and the statistically significant negative effect of the interaction between manufacturers' digital innovation and distributors' network status on distributors' digital innovation (b = −0.282, p = 0.003). Based on the results, all hypotheses are further supported.

Second, we used alternative measures for assessing our moderating variables – manufacturers' network status and distributors' network status. We used eigenvector centrality, an effective metric for assessing network influence and power, to measure network status (Kim and Zhu, 2018; Wang *et al.*, 2015). By taking into account both direct and indirect connections, eigenvector centrality plays a crucial role in understanding status within social networks (Lan *et al.*, 2020; Li *et al.*, 2023). Based on the ego networks of manufacturers and distributors, we obtained alternative manufacturers' network status and distributors' network status. The results are presented in Models 9 to 11 in Table 3. These models build upon the foundations of Models 1 and 2 from Table 2. Model 9 adds manufacturers' network status and the interaction term. Model 10 adds distributors' network status and the interaction term. Model 11 is the full model, which incorporates cooperation length, manufacturers' network status, distributors' network status and the interaction terms. Model 9 reveals the statistically significant positive effect of the interaction between manufacturers' digital innovation and manufacturers' network status on distributors' digital innovation (b = 0.488, p = 0.002). Model 10 indicates that the interaction of manufacturers' digital innovation and distributors' network status is statistically negative (b = −0.225, p = 0.028). Model 11 shows consistent results. All this further supports the findings of this study.

Finally, we used a modified sample to ensure robustness. While our sample was structured around pairs of manufacturers and distributors, it is important to note that the relationship between them is not strictly one-to-one. Specifically, a single distributor might be associated with multiple manufacturers and digital innovation from these various manufacturers could collectively influence the distributor, potentially introducing a slight bias into our findings. To mitigate this, we carefully selected only those pairs where a single distributor corresponds to a single manufacturer, rerunning our analyses on this refined data set. The modified sample encompassed 139 such unique pairs of manufacturers and distributors. Table 4 shows the results. Model 12 presents the base model on distributors' digital innovation. Model 13 shows the main effect of manufacturers' digital innovation on

Table 3 Robustness checks: alternative independent variable and moderate variables

Variables	Model 7	Model 8	Model 9	Model 10	Model 11
Manufacturers' firm age	0.006 (0.583)	0.010 (0.307)	0.004 (0.706)	0.004 (0.705)	0.003 (0.778)
Manufacturers' firm size	0.217† (0.054)	0.239* (0.028)	0.116 (0.328)	0.248* (0.021)	0.116 (0.304)
Distributors' firm age	−0.039** (0.003)	−0.033* (0.021)	−0.038** (0.003)	−0.042** (0.001)	−0.035** (0.008)
Distributors' firm size	0.258* (0.028)	0.282* (0.012)	0.246* (0.033)	0.247* (0.027)	0.264* (0.012)
Manufacturers' digital innovation	0.366** (0.001)	0.494*** (0.000)	0.555*** (0.000)	0.388*** (0.000)	0.683*** (0.000)
Cooperation length		−0.057 (0.677)			−0.004 (0.977)
Manufacturers' digital innovation * cooperation length		0.483** (0.001)			0.336** (0.007)
Manufacturers' network status		−0.149 (0.132)	−0.133 (0.241)		−0.195† (0.063)
Manufacturers' digital innovation * manufacturers' network status		0.390** (0.002)	0.488* (0.002)		0.509*** (0.000)
Distributors' network status		0.109 (0.240)		0.411*** (0.000)	0.440*** (0.000)
Manufacturers' digital innovation * distributors' network status		−0.282** (0.003)		−0.225* (0.028)	−0.234** (0.009)
Constant	−0.611 (0.539)	−0.939 (0.346)	0.295 (0.780)	−0.756 (0.420)	0.150 (0.881)
N	233	233	233	233	233
Log-likelihood	−544.578	−529.968	−538.477	−535.203	−521.012

Note(s): p-values are in parentheses. $^{\dagger}p < 0.10$; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$
Source(s): Authors' own work

Table 4 Robustness checks: modified sample

Variables	Model 12	Model 13	Model 14	Model 15	Model 16	Model 17
Manufacturers' firm age	0.011 (0.256)	0.017† (0.067)	0.016† (0.078)	0.015† (0.091)	0.017† (0.057)	0.015† (0.076)
Manufacturers' firm size	0.204* (0.049)	0.332** (0.001)	0.321** (0.002)	0.342** (0.001)	0.333** (0.001)	0.325** (0.001)
Distributors' firm age	−0.025† (0.054)	−0.027* (0.025)	−0.036** (0.008)	−0.027* (0.019)	−0.028* (0.017)	−0.039** (0.003)
Distributors' firm size	0.162 (0.120)	0.176† (0.074)	0.191† (0.052)	0.163† (0.080)	0.205* (0.034)	0.220* (0.016)
Manufacturers' digital innovation		0.364*** (0.000)	0.373*** (0.000)	0.349*** (0.000)	0.374*** (0.000)	0.367*** (0.000)
Cooperation length			0.152 (0.146)			0.178† (0.071)
Manufacturers' digital innovation * cooperation length			0.051 (0.575)			0.093 (0.271)
Manufacturers' network status				−0.253** (0.004)		−0.231** (0.006)
Manufacturers' digital innovation * manufacturers' network status				0.337** (0.003)		0.385** (0.001)
Distributors' network status					0.113 (0.180)	0.084 (0.285)
Manufacturers' digital innovation * distributors' network status					−0.213* (0.013)	−0.216** (0.006)
Constant	−0.061 (0.946)	−1.112 (0.208)	−0.938 (0.290)	−1.069 (0.221)	−1.228 (0.154)	−1.006 (0.237)
N	139	139	139	139	139	139
Log-likelihood	−395.006	−387.378	−386.157	−379.949	−383.739	−373.870

Note(s): p-values are in parentheses. $^{\dagger}p < 0.10$; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$
Source(s): Authors' own work

distributors' digital innovation. Models 14 to 16 incorporate cooperation length, manufacturers' network status and distributors' network status, respectively, along with their corresponding interaction terms. Model 17 is the full model. Specifically, in Model 13, manufacturers' digital innovation is positively associated with distributors' digital innovation ($b = 0.364$, $p = 0.000$). Model 14 showcases a positive but not significant impact of the interaction between manufacturers' digital innovation and cooperation length ($b = 0.051$, $p = 0.575$). Model 15 reveals the statistically significant positive effect of the interaction between manufacturers' digital innovation and manufacturers' network status on distributors' digital innovation ($b = 0.337$, $p = 0.003$). Model 16 indicates that the interaction of manufacturers' digital innovation and distributors' network status is statistically negative ($b = -0.213$, $p = 0.013$). Three of the four hypotheses were similarly supported. Model 17 demonstrates consistent results. The findings almost align consistently with the previously observed results.

Discussion

This study examines the diffusion of digital innovation in the manufacturer–distributor relationship. Drawing upon diffusion of innovation theory and social network theory, we develop a theoretical framework to analyze how digital innovation spreads from manufacturers to distributors. Specifically, we focus on the moderating roles of cooperation length, as well as the network status of manufacturers and distributors, in shaping the diffusion process. The results provide support for our theoretical predictions. Our findings underscore that manufacturers' digital innovation exerts a positive influence on distributors' digital innovation. Notably, cooperation length enhances the positive relationship between manufacturers' digital innovation and distributors' digital innovation. In contrast, while manufacturers' network status positively strengthens this relationship, distributors' network status appears to dampen the positive impact of manufacturers' digital innovation on distributors' digital innovation.

Theoretical contributions

This study makes several important theoretical contributions. First, this study offers a significant theoretical contribution by providing a nuanced understanding of how digital innovation diffuses from manufacturers to distributors within the supply chain. The existing literature on digital innovation primarily focuses on either the antecedents or the impacts of digital innovation, but there is a gap in our knowledge regarding how these advancements spread across organizations, particularly within the supply chain (Huo *et al.*, 2014; Kumar *et al.*, 2018). By examining the manufacturer–distributor relationship, this study fills this gap by identifying the mechanisms through which upstream firms influence downstream firms' adoption of digital innovation. This theoretical framework advances our comprehension of supply chain dynamics in the digital era. It underscores the critical role of upstream firms' digital innovation practices in shaping downstream firms' technological capabilities and market strategies. Specifically, the study reveals that manufacturers' digital innovation efforts often set the technological standards and operational paradigms for the entire supply chain, influencing distributors' adoption of digital innovation. Moreover, the alignment pressure between manufacturers and distributors ensures compatibility and interoperability, driving distributors to innovate digitally. This theoretical contribution offers valuable insights into how firms collaborate and learn from each other to drive innovation and growth within the supply chain, enriching existing theories on digital innovation and supply chain management.

Second, this study extends the application of social network theory to the context of digital innovation diffusion from manufacturers to distributors in the supply chains by incorporating cooperation length as a moderator. While prior research has examined the role of cooperation length in various contexts (Gao *et al.*, 2015; Han *et al.*, 2022), its specific impact on digital innovation diffusion remains underexplored. By highlighting the importance of relationship stability and trust in facilitating the exchange of knowledge and resources (Han *et al.*, 2022; Zhao *et al.*, 2008), this theoretical extension provides new insights into how cooperation length shapes the diffusion process of digital advancements across supply chain partners. Specifically, the study shows that longer cooperation periods foster stronger trust, familiarity and a shared understanding between manufacturers and distributors. This enhanced relationship facilitates the smooth transmission of digital innovation practices and technologies, making distributors more likely to adopt manufacturers' digital innovation. Moreover, longer cooperation lengths allow for mutual investment and support in relationship maintenance and enhancement, further strengthening the positive relationship between manufacturers' and distributors' digital innovation. This theoretical contribution offers a more nuanced understanding of the role of social relationships in digital innovation diffusion, extending existing theories on diffusion of innovation and supply chain collaboration.

Third, by investigating the moderating effects of manufacturers' and distributors' network positions within the context of social network theory, this study makes a significant theoretical contribution, greatly improving our comprehension of how firms' social influence impacts the diffusion of digital innovation throughout supply chains. While network status has been examined in the context of social networks and firm performance (Lan *et al.*, 2020; Li *et al.*, 2023), its specific impact on digital innovation diffusion has received limited attention. By drawing upon social network theory, the research examines how firms' strategic network positions influence their capacity to adopt and disseminate digital advancements. The theoretical integration of network status and digital innovation diffusion offers a comprehensive view of how network dynamics moderate the spread of digital innovation across supply chain partners. Specifically, the study reveals that manufacturers' high network status enhances their ability to disseminate digital innovation effectively and influence distributors' digital innovation. Conversely, distributors' high network status may lead to resistance to change and reluctance to follow manufacturers' lead in digital innovation due to their established business models and market positions. This theoretical contribution advances our understanding of the intersection between social networks and digital technologies.

Finally, this study also builds on and extends existing research by providing a more comprehensive understanding of the antecedents, diffusion process and impacts of digital innovation within the supply chains. While prior studies have focused on various factors that enable the successful implementation of digital innovation (Ferreira *et al.*, 2019; Van Looy, 2021) and its impact on competitive advantage, operational efficiency and market positioning (Fichman *et al.*, 2014; Porter and Heppelmann, 2014), this study bridges the gap between these two areas by exploring how digital innovation diffuses within the supply chains. By integrating diffusion of innovation theory, social network theory and supply chain management literature, this study offers a more holistic perspective on the role of digital innovation in driving business transformation and market adaptability.

Managerial implications

The findings of this study have several important implications for supply chain management across industries. First, this study underscores the importance of strategic alignment within the supply chains. Managers should prioritize forging partnerships with distributors who are willing and capable of adopting and adapting to new digital technologies and innovations. This alignment ensures that the benefits of digital innovation can be realized throughout the entire supply chain, from raw material sourcing to end-customer delivery, thereby enhancing overall supply chain efficiency and responsiveness.

Second, managers at manufacturing firms should recognize that their digital innovation efforts can positively influence their distributors, ultimately boosting the entire supply chain's competitiveness. By investing in cutting-edge digital technologies and solutions, manufacturers cannot only improve their own operational efficiency and market competitiveness but also inspire their distributors to follow suit. Managers should communicate the benefits of these innovations to distributors and provide the necessary support and resources to facilitate adoption.

Third, the positive moderating effect of cooperation length highlights the importance of fostering long-term and stable relationships with distributors. Managers should invest in relationship maintenance and enhancement activities, such as shared resources, training programs and joint innovation initiatives. These investments not only strengthen the partnership but also equip distributors with the necessary capabilities and resources to adopt and implement new digital technologies and practices, fostering a collaborative and resilient supply chain.

Fourth, manufacturers occupying central network positions have greater access to resources, information and influence, which can enhance their ability to disseminate digital innovation effectively. Managers should strive to position their firms strategically within their networks to leverage these advantages. This may involve identifying and cultivating key relationships with partners who occupy central positions and can facilitate the diffusion of digital innovation, thereby optimizing the flow of information and resources within the supply chain.

Finally, managers should be cognizant of the fact that distributors' network status can impact their willingness and ability to adopt digital innovation. High-status distributors may require additional incentives and encouragement to overcome barriers to adoption, whereas low-status distributors may benefit from supplementary support and resources. Managers should tailor their diffusion strategies accordingly to maximize the impact of digital innovation across the supply chain, ensuring that all participants can benefit from and contribute to a more digitalized and efficient supply chain management system.

Limitations and future research

This study, while providing valuable insights into the diffusion of digital innovation in the manufacturer–distributor relationship, is not without limitations that offer avenues for future research. First, the focus on the Chinese paper-making industry limits the generalizability of the findings to other industries. Future studies could replicate this research in diverse industries and across different countries to further test the applicability and generalizability of the findings. This would not only provide a more comprehensive understanding of digital innovation diffusion across supply chains but also assess whether the results hold true in various cultural, economic and institutional contexts. Second, while several important variables were considered, there may be other factors that influence the diffusion of digital innovation, such as cultural differences, geographic proximity and regulatory environments. Future research could explore the role of these additional variables. Third, this study relied on questionnaire surveys and archival data. Conducting qualitative studies, such as in-depth interviews or case studies, could provide further insights into the mechanisms and dynamics of digital innovation diffusion across supply chain partners. This could complement the quantitative findings presented in this study.

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